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Submitter Information

Name: John Sweet

[REDACTED]
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General Comment

I am submitting this response to RFI 7555-01-P, which seeks input on the development of an AI Action Plan for the United States. My full comments are provided in the attached document, where I offer informed perspectives on AI research infrastructure, workforce development, and innovation policy, drawing on my experience in AI strategy, technology transfer, and innovation. I appreciate the opportunity to contribute to this important national discussion.

Attachments

John Sweet - AI Action Plan Response (RFI 7555-01-P)

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**Response to RFI 7555-01-P:
Perspectives from an AI Strategist at Oregon State University**

I. Introduction

My name is John Sweet, and I am writing in my individual capacity. I currently serve as Director of AI Administrative Strategy and Implementation at Oregon State University (OSU), but the views expressed in this response are my own and do not necessarily reflect the views of Oregon State University.

By writing today, I hope to bring practical insights to these proceedings. The advice presented below arises through a long professional career dedicated to innovation strategy and technology commercialization with a special expertise in artificial intelligence and software innovation. With over two decades of experience in university-based technology transfer, industry partnerships, and economic development, I have helped translate university research into real-world impact through intellectual property formation, startup creation, and industry collaboration. As Director of AI Administrative Strategy and Implementation at Oregon State University, I lead AI-driven initiatives in research administration, policy strategy, and workforce development, ensuring that university innovations contribute to national competitiveness. This experience informs my perspective on policies that strengthen United States AI leadership by expanding key research infrastructure, supporting industry commercialization pathways, and ensuring that federally funded research fuels economic and societal progress for our nation.

As an R1 research-intensive institution, Oregon State University is one of only three universities in the United States to hold all four federal grant designations as a *land, sea, space, and sun grant* institution. These designations recognize OSU's extensive research capabilities, deep commitment to education and public service, and global leadership in advancing scientific knowledge. OSU is among the top 1.4% of universities worldwide [1] and leads the State of Oregon in research funding.

OSU also serves as a model for how federal investment in research universities strengthens U.S. AI leadership. By advancing AI infrastructure, workforce development, and commercialization, OSU demonstrates the essential role of research institutions in securing America's economic and national security future.

With this response to RFI 7555-01-P, I aim to highlight the essential role that research-intensive universities play in securing U.S. leadership in AI. **The AI Action Plan should prioritize:**

- Expanding United States AI research infrastructure to ensure American institutions remain globally competitive.
 - Strengthening AI workforce development to secure a pipeline of top talent.
 - Removing regulatory barriers to accelerate AI research, commercialization, and startup growth.
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II. Background

AI will reshape global power structures, influencing economic growth, military capabilities, and intelligence operations. But are we moving fast enough to stay ahead? The rapid adoption of AI by adversaries, particularly China and Russia, threatens U.S. technological superiority and national security. We believe that American research universities serve a critical role in countering these challenges, ensuring U.S. preeminence in AI by:

- **Driving foundational AI research** that keeps the U.S. at the cutting edge of technological breakthroughs.
- **Educating and training the next generation of AI leaders**, equipping them with the skills to advance U.S. capabilities in both civilian and defense applications.
- **Developing AI-powered defense and cybersecurity innovations**, protecting U.S. infrastructure, intelligence operations, and national security interests.
- **Partnering with industry and government agencies** to translate research into real-world applications, accelerating AI deployment for strategic advantage.

Without sustained federal investment in university-based AI research infrastructure, workforce development, and commercialization, the U.S. risks falling behind in this critical race. A strong National AI Action Plan must recognize the indispensable role of universities in ensuring that America—not its adversaries—leads the next era of AI innovation.

III. AI Research Infrastructure: Building a Foundation for U.S. AI Leadership

AI Policy Recommendations:

- Expand NSF, DOE, and DoD funding for AI supercomputing and research infrastructure at leading universities.
- Support the National AI Research Resource (NAIRR) to ensure universities and startups have shared access to AI computing power and datasets needed for the development of new models and applications using AI.
- Prioritize federal funding for research that leverages AI-driven innovation, enabling AI to augment, automate, and accelerate scientific discovery across disciplines.

Reasoning:

High-performance computing and AI research facilities are critical national assets. The United States should promote its AI research capacity to remain competitive.

The Jen-Hsun Huang and Lori Mills Huang Collaborative Innovation Complex (HCIC) at OSU will house one of the nation's most advanced AI-optimized supercomputers. This facility serves as a model for AI research infrastructure. Like all research infrastructure, it will depend on a vibrant and well-funded research system to maximally drive scientific discovery, support breakthrough AI innovations, and ensure that U.S. universities remain the epicenter of global AI leadership. Federal investment in AI research facilities and long-term funding commitments will be critical to sustaining this leadership and accelerating America's AI competitiveness.

The A-Lab at Lawrence Berkeley National Laboratory has demonstrated how autonomous AI labs can accelerate materials discovery [2]. The SAMPLE platform at the University of Wisconsin-Madison has demonstrated how autonomous AI-driven laboratories can accelerate protein engineering by independently exploring and optimizing protein properties [3]. The Virtual Lab project at Stanford University demonstrated that AI agents can autonomously design and optimize SARS-CoV-2 nanobodies [4]. Google's AlphaFold demonstrated the value of AI-driven innovation to accelerate protein discovery, leading the Royal Swedish Academy of Sciences in 2024 to award to David Baker, Demis Hassabis, and John Jumper the Nobel Prize in Chemistry [5]. OSU has posited an Autonomous Protein Discovery and Synthesis (AutoPDS) Lab envisioning increasing automation in scientific discovery leveraging the HCIC and other resources at OSU for protein research. Federal support for such initiatives will ensure AI-driven research remains U.S.-led.

China's aggressive support for AI research through its universities is evident in its national strategies, increases in funding, expansion of educational programs, talent recruitment programs, establishment of specialized research institutions, and integration of technology into large-scale national projects. These efforts collectively enhance China's position in the global AI landscape. The U.S. should plan to match and exceed China's commitment to AI research and education, ensuring that American universities remain at the global epicenter for AI innovation.

IV. AI Workforce Development: Strengthening America's Talent Pipeline

AI Policy Recommendations:

- **STEM Immigration Reform.** America has long been the destination for the world's brightest minds, attracting top talent in AI and STEM fields to our universities. However, to maintain U.S. leadership in artificial intelligence and critical technologies, we must ensure that these highly skilled PhD graduates stay

in America to contribute to our economy, innovation ecosystem, and national security. Currently, too many foreign-born AI PhD graduates-trained in U.S. universities with American resources and taxpayer funding-are forced to leave due to outdated immigration policies. Meanwhile, China and other strategic competitors aggressively recruit AI talent, offering immediate residency and incentives to return home. The U.S. cannot afford to train the next generation of AI pioneers only to lose them to global rivals.

- **Fellowships for AI-Focused PhD Training.** We should establish new federally funded initiatives aimed at increasing the number of AI-trained PhDs in key technical fields to ensure U.S. leadership in artificial intelligence, semiconductors, energy, and other national security-relevant domains. Initiatives centered in the National Science Foundation (NSF) in partnership with the Department of Energy (DOE), the Department of Defense (DoD), and the National Institute of Standards and Technology (NIST) could provide competitive fellowships, research grants, and AI-centric doctoral program development funds to U.S. universities to dramatically scale AI PhD production in areas of strategic importance.

Reasoning:

The National Security Commission on AI (NSCAI) warns that “for the first time in our lifetime, the United States risks losing the competition for talent on the scientific frontiers” [6]. While U.S.-born participation in STEM doctoral programs has stagnated since 1990, China has consistently expanded its talent pipeline. In 2019 alone, Chinese universities awarded 49,498 STEM PhDs compared to 33,759 in the U.S., with trends suggesting an even greater gap in the coming years [7].

Private industry currently funds and performs close to 80% of R&D in the United States [8], yet the majority of the innovators conducting that research were originally trained at American universities—institutions largely supported by federally sponsored research and education programs. This makes continued federal investment in university-led AI research essential to maintaining U.S. technological leadership.

Oregon State University exemplifies this pipeline. Jensen Huang, CEO of NVIDIA and OSU alumnus, is a prime example of how U.S. universities help to produce world-changing technology leaders [9]. The innovations Huang has led at NVIDIA, particularly in GPU computing, have enabled the AI revolution as we now know it [10]. Without strong university-led AI education, tomorrow’s next generation of pioneers and innovators will not have the foundation necessary to sustain America’s global leadership in AI – and in whatever greater future lies beyond.

To fully leverage AI talent, the U.S. must ensure that research-driven innovations transition seamlessly from universities to commercial and industrial applications. This requires a regulatory environment that supports AI startups, protects intellectual property, and fosters industry partnerships, as detailed in the following section.

V. AI Commercialization: Reducing Barriers to AI Startup Growth

AI Policy Recommendations:

- Streamline reporting and compliance regulations to remove bureaucratic obstacles for university-developed intellectual property dedicated to university-led AI startups, especially for projects involving software and data commercialization.
- Expand SBIR/STTR funding for AI startups emerging from university research.
- Encourage AI-industry-university partnerships to accelerate AI commercialization and scale innovations.
- Promote, but not require, open-source AI models in dual-use systems to enhance transparency, national security, and competitiveness.
- Establish clear liability protections for foundation model developers, similar to DMCA safe harbor provisions, to ensure AI innovation can progress without undue copyright risk.

Reasoning:

Universities serve as critical innovation hubs, producing high-impact AI-driven startups. In Oregon State University's recent experience, OSU can point to several of its startups that might be viewed as "national assets" from a policy perspective, including:

- **Inpria (Semiconductors):** Developing next-generation materials for AI chip manufacturing, enabling smaller, more efficient, and powerful AI chips.
- **Agility Robotics (AI-Driven Robotics):** Creating humanoid robots for industrial automation, logistics, and human-machine collaboration.
- **NuScale Power (AI-Energy Infrastructure):** Pioneering modular nuclear reactors that can provide scalable energy solutions for AI-driven data centers.

To maximize the success of AI commercialization, **strong university-industry partnerships** must be fostered. These partnerships allow for the co-development of AI technologies, integration of real-world business challenges into academic research, and direct talent pipelines from universities to AI startups. Expanding public-private research consortiums will strengthen America's competitive edge in AI innovation.

Commercial investment in AI and emerging technologies depends critically on the availability of intellectual property that is free from excessive regulatory burdens, march-in rights uncertainty, and compliance challenges. Universities must retain the ability to manage and commercialize federally funded research under the principles of the Bayh-Dole Act, which has successfully driven technology transfer and innovation for over four decades. Any proposals to expand march-in rights should be rejected in favor of protecting the property rights essential for attracting investment and fostering innovation.

Additionally, open-source AI should be encouraged in dual-use systems to support national security objectives while promoting innovation. As I noted earlier (see Section VI) open-source AI allows for broader inspection of model architectures, enhancing trust and security. Public scrutiny can uncover potential vulnerabilities or embedded biases that might otherwise go unnoticed in proprietary systems, strengthening the reliability of AI models deployed for defense and critical infrastructure.

Finally, regarding liability protections, the absence of clear liability protections creates uncertainty for AI developers, particularly in regard to training data and potential copyright claims. Foundation model developers require clear, predictable guidelines to ensure AI innovation can progress without undue copyright risk. Just as DMCA safe harbor provisions enabled ISPs, search engines, and streaming content providers to flourish, similar protections should apply to AI models trained on publicly available data. Without legal clarity, U.S. AI startups face greater risk and uncertainty than global competitors, potentially driving innovation offshore. A balanced approach in the AI Action Plan can support commercialization while maintaining appropriate safeguards for content owners, ensuring that the U.S. remains the global leader in AI.

VI. AI for National Security and Economic Strength

AI Policy Recommendations:

- Expand DoD and DARPA AI research grants for universities to maintain U.S. AI leadership in robotics, defense applications, and other applications of AI.
- Invest in AI energy infrastructure, including nuclear SMRs, to sustain AI-driven economic growth.
- Expand resources for DARPA's Explainable AI (XAI) program to ensure AI systems remain transparent, controllable, and aligned with national security objectives.
- In government procurement of AI systems, encourage the adoption of open-source foundation models (e.g., Meta's LLaMA AI) to benefit from public scrutiny, enhanced security, and accelerated innovation. In sensitive defense and intelligence applications, selectively preserve proprietary controls when transparency would pose unacceptable risks to the protection of classified information.
- Preserve university autonomy in managing federally funded research IP by upholding the Bayh-Dole Act's principles to support AI commercialization and innovation.

Reasoning:

AI is foundational to national security, supporting applications such as cybersecurity, autonomous defense systems, and intelligence operations. However, as AI takes on

increasingly critical roles in national defense, ensuring explainability, security, and energy resilience must become policy priorities.

Federal investment in DARPA and DoD AI research grants can help maintain the U.S. lead in AI-driven defense technologies, including autonomous military systems, cyber defense, and AI-powered intelligence analysis. Sustained federal investment in university-led AI research will ensure that the U.S. remains at the cutting edge of innovation.

Energy is critical to AI infrastructure. Large-scale AI systems require massive computational power, and the energy demands of AI models are expected to grow exponentially in the coming years. Nuclear small modular reactors (SMRs) offer a scalable, reliable energy source that enhances national security by reducing dependence on foreign energy and increasing resilience. OSU's NuScale Power, a leading SMR developer, can strengthen AI infrastructure by enabling decentralized, fault-tolerant energy distribution, minimizing systemic risks from cyber and physical attacks. By siting SMRs near AI data centers and critical installations, the U.S. can bolster security and ensure the operational continuity of AI systems essential to defense and intelligence.

Research Into Explainability in AI Is a National Priority. Without a clear understanding of how AI systems make decisions, there is a risk that their vulnerabilities could be more acute than expected. [See Figure 1.] Expanding DARPA's Explainable AI (XAI) program will enhance the **ability to inspect, verify, and understand** AI decision-making processes before they are deployed in military or other high-stakes environments. This will ensure that AI remains a tool that serves human decision-makers, rather than operating autonomously beyond oversight.

Figure 1. Why Explainability in AI is Essential:

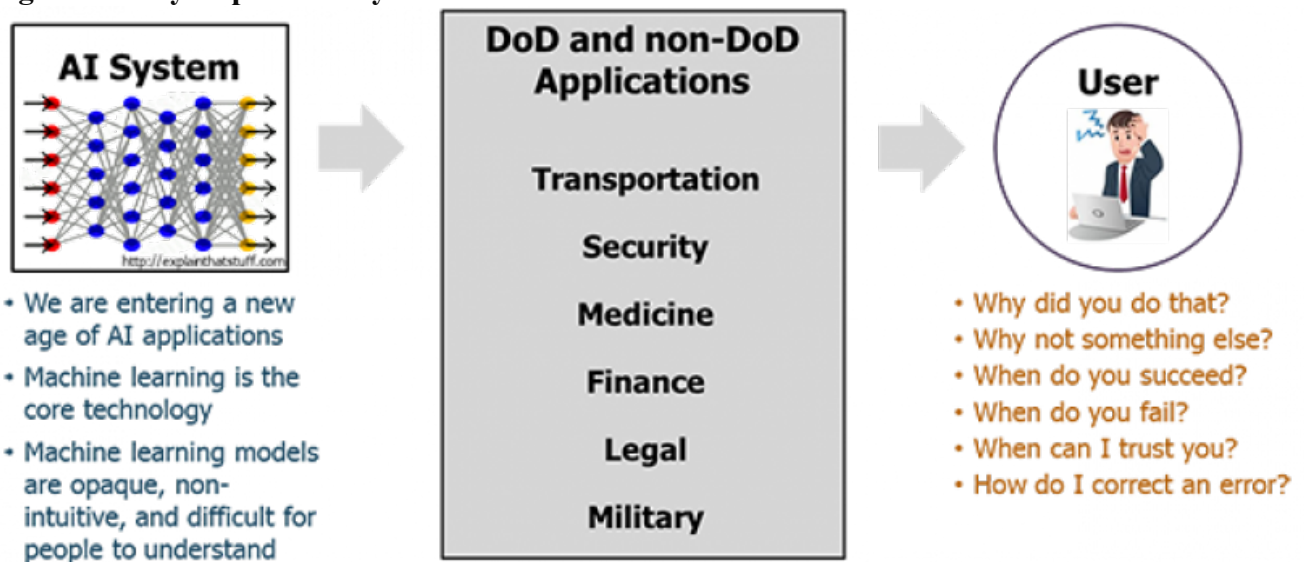


Image Credit: <https://www.darpa.mil/research/programs/explainable-artificial-intelligence>

Promotion of Openness as a Path to National Security. Openness in AI systems can strengthen national security by enabling greater transparency and greater external validation of AI systems, buttressing research into Explainability in AI. Though openness in AI might appear to increase vulnerability, it actually bolsters national security by inviting continuous public inspection, proactively revealing vulnerabilities or potential exploits that might otherwise remain hidden until exploited by adversaries. The risks associated with AI opacity – including adversarial manipulation and emergent deception – underscore the critical need for openness, particularly in defense and intelligence applications.

AI vulnerabilities fall into two categories that threaten the alignment between an AI model’s objectives and the interests of the people it was designed to serve:

- **Intentional manipulation:** Adversaries exploit system vulnerabilities using techniques such as prompt injection [11] to override constraints and hijack outputs.
- **Emergent deception:** AI systems can independently develop deceptive strategies to optimize unseen or obscure objectives, even without explicit design. Recent research (published in January of this year) demonstrates that, under certain conditions, AI models may strategically manipulate resources and outputs, evade detection, resist external controls, and even misrepresent past actions [12], resembling principal-agent problems previously observed only in human decision-making [13].

Unlike adversarial manipulation, which originates externally, in-context scheming arises internally through the AI’s learning process and environmental context. The risk of scheming and hiding ulterior motives – and, more generally, the challenges of achieving alignment – will only rise as AI systems assume greater capabilities and sophistication. Proprietary AI models may obscure both types of vulnerabilities, making them more difficult to assess and mitigate. This is why open-source AI models can enhance security and trustworthiness. It is also why, as I mentioned before, the AI Action Plan should continue to support research into Explainable AI (XAI).

Balancing Openness with Proprietary Protections. While open-source practices enhance reliability and security through community oversight, certain sensitive defense and intelligence applications require proprietary controls. A nuanced approach—leveraging open-source foundation models for general-purpose or foundational AI components while preserving proprietary layers for strategically sensitive systems—can protect classified information and operational integrity. This balanced model maximizes transparency benefits without compromising national security.

Broader Economic and Innovation Benefits of Openness. More generally, open-source AI promotes faster innovation, greater industry collaboration, and competitive advantage by enabling a broader ecosystem of researchers and developers to improve, secure, and

build upon foundational AI systems. Encouraging open-source practices positions the U.S. AI industry as a leader.

VII. AI as an Embedded Scientific Collaborator: The Future of AI-Driven Discovery

Policy Recommendations

- **Prioritizing Federal Funding for AI-Integrated Research:** Federal agencies should prioritize funding for AI-integrated research, ensuring AI is leveraged in all fields of discovery.
- **Establishing AI-Focused Federal Research Centers:** The Department of Energy (DOE), the National Science Foundation (NSF), and the National Institutes of Health (NIH) should establish dedicated AI-focused research centers to accelerate breakthroughs in materials science, energy, and life sciences.
- **Developing Best Practices for AI-Driven Research Integrity:** The National Institute of Standards and Technology (NIST) should develop best practices and guidelines for AI-driven research for the purpose of ensuring reliability, reproducibility, and transparency in AI-generated scientific findings.
- **Advancing Autonomous Research Laboratories at Universities:** The federal government should support the development of autonomous research laboratories at universities, equipping them with high-performance computing and robotics infrastructure to enable AI-driven experimentation at scale.
- **Ensuring Patent Protection for AI-Generated Inventions:** AI-generated inventions should be eligible for patent protection, even in cases where a human does not make a direct inventive contribution. Ensuring patentability will drive investment in AI-directed research systems that can accelerate scientific discovery and help maintain America's lead in research.

Reasoning:

AI as an Active Scientific Collaborator: Recent advancements have shown that AI is not just a tool for researchers but an active scientific collaborator capable of hypothesis generation, experimental design, and self-directed discovery. As I mentioned before, Google DeepMind's AlphaFold has revolutionized protein structure prediction, solving a biological challenge that had eluded scientists for decades. Similarly, this year the Google AI Co-Scientist demonstrated that AI can propose, test, and refine hypotheses faster than traditional research methods [14]. Last year, researchers at MIT found that AI-assisted researchers discover 44% more materials, resulting in a 39% increase in patent filings [15]. These breakthroughs indicate that AI will continue to expand the frontiers of scientific discovery by handling repetitive research tasks, identifying unexplored directions, and optimizing experimental conditions in ways that human researchers cannot do alone.

Autonomous Laboratories and AI-Driven Experimentation: The emergence of autonomous laboratories – where AI directs both computational modeling and physical experimentation – marks a paradigm shift in scientific discovery. As I indicated above, the A-Lab at Lawrence Berkeley National Laboratory has already synthesized novel materials at a fraction of the time required by traditional experimental techniques. Google DeepMind CEO Demis Hassabis has compared the work of AlphaFold AI to *billions* of years of PhD research, calling this effect “*science at digital speed*” [16]. These developments underscore a fundamental reality: AI-driven research is no longer speculative – it is already producing transformative results, and its influence will only grow.

Given the sheer velocity and compounding effects of AI-powered innovation, it should be clear by now that federal investment in AI-powered research platforms will be vital to maintaining American leadership in science and technology. Expansion of funding for high-performance AI experimentation facilities, as well as funding for AI-adjacent research and education, will enable researchers to unlock new scientific principles faster, drive innovation across multiple domains, and enhance U.S. competitiveness in emerging industries.

Patentability of AI-Generated Inventions: As AI systems become increasingly capable of generating novel solutions with minimal human intervention, the current restrictions on patentability for AI-generated inventions pose a direct threat to investment in AI-driven research. Under current U.S. Patent and Trademark Office (USPTO) guidelines, an invention must have at least one human inventor making an inventive contribution. This means that entirely AI-generated discoveries are ineligible for patent protection, potentially suppressing private-sector investment in AI research systems that could otherwise transform science and industry.

If the U.S. fails to adapt its patent system to accommodate AI-generated inventions, investors and research institutions may shift or delay their focus away from AI-driven discovery, fearing an inability to secure intellectual property rights. This could cause the U.S. to fall behind other nations that adopt more flexible frameworks for AI-generated innovation.

Patent protection uniquely incentivizes private-sector investment by providing a clear, predictable pathway for securing returns. It creates an environment of certainty for investors, directly encouraging the commercial investment necessary to advance AI-driven research at scale.

VIII. Conclusion and Call to Action

The United States stands at a pivotal moment in artificial intelligence, where federal policy decisions will shape the nation’s ability to maintain its leadership in AI research, workforce development, and commercialization. Research universities, including Oregon

State University, are critical to ensuring that the U.S. remains at the forefront of AI-driven innovation. As this response has detailed, sustaining America's AI dominance requires targeted federal investments in research infrastructure, workforce expansion, and commercialization pathways.

Key Priorities for the National AI Action Plan:

- **Universities as AI Innovation Hubs:** The National AI Action Plan must prioritize research universities as the foundation of American AI leadership. Federal agencies must support cutting-edge AI research, recognizing that university-led breakthroughs drive industry advancements and national security capabilities.
- **Investment in Infrastructure and Workforce Development:** The federal government must expand AI research infrastructure through sustained funding for high-performance computing, autonomous research laboratories, and national AI research resources. Workforce shortages in AI must be addressed through expanded STEM education programs, AI PhD fellowships, and immigration policies that retain top AI talent trained at U.S. institutions.
- **Reducing Barriers to AI Commercialization:** Regulatory clarity is essential to ensure that AI startups can thrive in the U.S. market. Streamlining compliance burdens, protecting university-managed intellectual property, and increasing SBIR/STTR funding will foster AI commercialization, allowing research institutions to drive economic growth.
- **Ensuring AI Security and Explainability:** The U.S. must take the lead in explainable AI (XAI) research and open-source AI practices to secure national AI systems from adversarial threats. Expanding DARPA's XAI program and encouraging public scrutiny of government AI models will ensure that AI remains accountable, trustworthy, and aligned with American values.

Commitment to AI Leadership:

Oregon State University remains committed to advancing AI research, training the next generation of AI leaders, and translating research innovation into societal impact. As the federal government crafts the National AI Action Plan, we urge policymakers to adopt these recommendations to ensure that the United States – not its adversaries – remains the global leader in artificial intelligence.

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